

- 1) What is the minimum and maximum size of a UDP datagram?
- 2) What is the minimum and maximum size of the process data that can be encapsulated in a UDP datagram?
- 3) What is the maximum size of the TCP header? What is the minimum size of the TCP Header?
- 4) The following is a dump of a UDP header in hexadecimal format.
0632 000D 001C E217
 - a. What is the source port number?
 - b. What is the destination port number?
 - c. What is the total length of the user datagram?
 - d. What is the length of the data?
 - e. Is the packet directed from a client to a server or vice versa?
 - f. What is the client process?
- 5) Why do we need a DNS system when we can directly use an IP address?
- 6) In Symmetric-key cryptography, can Alice use the same key to communicate with both Bob and John? Explain your answer.
- 7) In Asymmetric-key cryptography, can Alice use the same key to communicate with both Bob and John? Explain your answer.
- 8) A hash algorithm creates a digest of N bits. How many different digests can be created from this algorithm?
- 9) Explain the difference between a stateless firewall and a stateful firewall.